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Ekman Spirals with Variable Eddy Viscosity

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Contents

1 Introduction

The Ekman spiral is a famous consequence of the Coriolis force in meteorology and oceanography. Due to the interaction between shear forces in the atmosphere and ocean, the wind high up in the atmosphere, and wind over the ocean, will not give rise to a flow in the same direction at the surface. The adjustment of the flow from the surface will thus spiral, and is therefore properly named the Ekman spiral after its discoverer Ekman [1905]. The surface flow is often at an angle from the flow aloft by an angle of more or less 45° [Ekman, 1905]. Due to this turning of the wind, the net flux of the wind will be perpendicular to the wind aloft. This is known as the Ekman transport and is crucial for studying both meteorology and oceanography.

However, the wind turning 45° assumes a constant viscosity in the medium. This is unrealistic when a turbulent boundary layer is present. The boundary layer is around 1–2 km thick in the atmosphere and is represented by the presence of turbulence [Wallace and Hobbs, 2006]. This turbulence arises primarily from solar heating, causing instability and turnover, but also from mechanical turbulence from the mean wind decreasing closer to the ground due to shear force caused by frictional drag. Although turbulence is non-linear, statistical analysis can provide useful first-order approximations, such as the downward flux being negatively and linearly proportional to the local gradient. That is, one cd the sheaan define the eddy viscosity, ν, for the boundary layer.

According to the mixing length hypothesis, this eddy viscosity should grow with the distance from the surface as the characteristic scales of the eddies grow. This can furthermore be described by a dimensional analysis of the boundary layer, where a logarithmic relation between the wind speed and height can be obtained [Wallace and Hobbs, 2006]. This showcases that the eddies, and thus the eddy viscosity, grow with height in the boundary layer. However, after a certain altitude, the turbulent boundary layer ceases to exist, and laminar flow takes over, which decreases the eddy viscosity. One can thus assume that the eddy viscosity should increase initially from the surface and then die down at a higher altitude.

Thus, in reality the viscosity should vary with altitude. This would change the wind-turning from 45°, and thus also the Ekman transport. In this article, a varying viscosity with altitude will be studied under the Ekman equation. This will be done both analytically and numerically and will also be compared to model data. This will give a deeper insight into the Ekman spiral and provide insight into when a surface angle different from 45° can be observed.

2 Theory

2.1 Problem formulation

The Ekman equation is derived from the steady-state Navier–Stokes equations on a rotating frame with shear flow. Assuming friction, incompressible flow, negligible vertical motion, gives the horizontal momentum equations as

$$\begin{cases} -fv = -\dfrac{1}{\rho}\dfrac{\mathrm{d}p}{\mathrm{d}x} + \dfrac{\mathrm{d}}{\mathrm{d}z}\left(\nu\dfrac{\mathrm{d}u}{\mathrm{d}z}\right), & \text{(1a)} \\ fu = -\dfrac{1}{\rho}\dfrac{\mathrm{d}p}{\mathrm{d}y} + \dfrac{\mathrm{d}}{\mathrm{d}z}\left(\nu\dfrac{\mathrm{d}v}{\mathrm{d}z}\right), & \text{(1b)} \end{cases}$$

where u and v are the horizontal velocity components, f is the Coriolis parameter, ρ is the fluid density, and ν is the viscosity. Far away from the surface, shear flow and friction becomes negligible and the flow will approach geostrophic balance, $(u, v) \to (u_g, v_g)$, where the friction term in Equation 1b can be neglected. Subtracting the geostrophic balance from Equation 1b gives

$$\begin{cases} -f(v - v_g) = \dfrac{\mathrm{d}}{\mathrm{d}z}\left(\nu\dfrac{\mathrm{d}u}{\mathrm{d}z}\right), & \text{(2a)} \\ f(u - u_g) = \dfrac{\mathrm{d}}{\mathrm{d}z}\left(\nu\dfrac{\mathrm{d}v}{\mathrm{d}z}\right). & \text{(2b)} \end{cases}$$

This problem can be formulated with one equation by introducing a complex velocity as

$$U \equiv u + iv, \tag{3}$$

and a corresponding complex geostrophic velocity. Using Equation 2b with the complex velocities U and U_g the coupled equations can be combined into the Ekman equation

$$\frac{\mathrm{d}}{\mathrm{d}z}\left(\nu\frac{\mathrm{d}U}{\mathrm{d}z}\right) = if(U - U_g), \tag{4}$$

with the boundary conditions

$$U(0) = 0, \tag{5a}$$

$$U(z \to \infty) = U_g. \tag{5b}$$

The surface wind-turning angle can be determined by taking the argument of the infinitesimal change of the wind at the ground,

$$\theta = \arg\left(\left.\frac{\mathrm{d}U}{\mathrm{d}z}\right|_{z=0}\right). \tag{6}$$

The Ekman transport is defined as the mass flux per unit width for the entire Ekman spiral, which can be defined as

$$T = \int_0^\infty [U - U_g]\,\mathrm{d}z, \tag{7}$$

since the transport represents the net movement of the entire fluid column, offset from the geostrophic wind aloft. The angle of the Ekman transport can consequently be determined by taking the argument, $\theta_T = \arg(T)$.

2.2 The vertical momentum flux

The Ekman equation can be reformulated by introducing a complex vertical momentum flux as

$$\tau = \nu\frac{\mathrm{d}U}{\mathrm{d}z}. \tag{8}$$

Using this definition, Equation 4 can be rewritten as

$$\boxed{\frac{\mathrm{d}^2\tau}{\mathrm{d}z^2} = i\underbrace{\lambda}_{\equiv\frac{f}{\nu}}\tau.} \tag{9}$$

Here, an eigenvalue λ has been introduced to the system. The new eigenvalue form is more intuitive since it directly compares the second order derivative with the variable itself. Furthermore, Equation 9 is comparable with famous equations such as the Helmholtz equation and the time-independent Schrödinger equation, where λ becomes analogous to the potential. The new equation also has a clear physical interpretation, as it describes the vertical structure of the momentum flux in the Ekman problem. Using Equation 9, the boundary conditions become

$$\left.\frac{\mathrm{d}\tau}{\mathrm{d}z}\right|_{z=0} = -ifU_g, \tag{10a}$$

$$\tau(z \to \infty) = 0. \tag{10b}$$

This new quantity will not change the surface angle, as the viscosity is always a positive real value and does not affect the argument,

$$\theta = \arg\left[\tau(z=0)\right]. \tag{11}$$

Solving for the Ekman transport becomes easier by using the momentum flux, as it directly reduces the integral of the Ekman transport. Observing the Ekman equation (Equation 4) it follows that $\frac{\mathrm{d}\tau}{\mathrm{d}z} = -if\left(U - U_g\right)$. Using the boundary condition that the momentum flux vanishes as infinity the Ekman transport reduces to

$$T = \int_0^\infty [U - U_g]\,\mathrm{d}z = \frac{i}{f}\tau(z=0), \tag{12}$$

which is proportional to the vertical momentum flux at the surface. Consequently, this can be interpreted as the surface stress at the ground being proportional to the Ekman transport of the fluid column.

2.3 The classical solution

The classical solution to the Ekman equation is obtained when the viscosity is assumed to be constant with altitude, ν = constant, and was found by Ekman [1905]. In this case the classical Ekman spiral solution is

$$\tau(z) = \nu U_g \frac{1+i}{h_{\mathrm{Ek}}} e^{-\frac{1+i}{h_{\mathrm{Ek}}} z}, \tag{13}$$

where

$$h_{\mathrm{Ek}} \equiv \sqrt{\frac{2\nu}{f}} \tag{14}$$

is the Ekman height scale. The surface angle of the classical Ekman theory with constant viscosity gives 45°, since the solution is equally real and imaginary at $z = 0$. The transport yields an angle of 90° from the surface wind-turing, meaning that the transport and surface wind is perpendicular.

2.4 The two layer model

The next natural step from a constant viscosity profile is to assume two layers of constant viscosity in each layer. One can label the lower level as having the viscosity ν_1 up until a height of H, where the next viscosity of ν_2 takes over. From this assumption a superimposed solution similar to Equation 13 is obtained as

$$\begin{cases} \tau_1(z) = A_1 e^{\frac{1+i}{h_{\mathrm{Ek1}}} z} + B_1 e^{-\frac{1+i}{h_{\mathrm{Ek1}}} z}, & \text{(15a)} \\ \tau_2(z) = B_2 e^{-\frac{1+i}{h_{\mathrm{Ek2}}} z} & \text{(15b)} \end{cases}$$

where the Ekman height scales are defined for each layer like in Equation 14. To solve for the prefactors above one has to use that the velocity and momentum flux is continuous, which gives the jump condition

$$U_1 = U_2 \quad \text{at } z = H, \tag{16a}$$

$$\tau_1 = \tau_2 \quad \text{at } z = H. \tag{16b}$$

Using the jump conditions together with the boundary conditions at Equation 5 an expression between the prefactors is given as

$$A_1 = -\nu_1 U_g \frac{1+i}{h_{\mathrm{Ek1}}} \cdot \frac{1-\gamma}{1-\gamma+e^{2(1+i)\varphi}(1+\gamma)}, \tag{17a}$$

$$B_1 = \nu_1 U_g \frac{1+i}{h_{\mathrm{Ek1}}} \cdot \frac{(1+\gamma)\, e^{2(1+i)\varphi}}{1-\gamma+e^{2(1+i)\varphi}(1+\gamma)}, \tag{17b}$$

$$B_2 = \nu_2 U_g \frac{1+i}{h_{\mathrm{Ek2}}} \cdot \frac{2e^{(1+i)\varphi\left(1+\gamma^{-1}\right)}}{1-\gamma+e^{2(1+i)\varphi}(1+\gamma)}, \tag{17c}$$

$$\text{where } \gamma = \sqrt{\tfrac{\nu_2}{\nu_1}} \tag{17d}$$

where φ is the non-dimensional layer thickness defined as

$$\varphi = \frac{H}{h_{\mathrm{Ek1}}}. \tag{18}$$

This provides a complete expression for the vertical momentum flux in the two-layer model, as shown in Figure 1. The figure illustrates how the vertical momentum flux varies with altitude for different viscosity ratios and non-dimensional heights, φ. For an increasing viscosity with height, a larger surface stress in the zonal direction is observed. Conversely, for a smaller upper-layer viscosity, a larger meridional surface stress is evident. Observing the interface, H, reveals that for larger φ, the model behaves more like a single-layer model, as the second layer begins at a height where the Ekman spiral has already decayed. Interestingly, smaller values of φ yield a larger difference between the zonal and meridional wind stress.

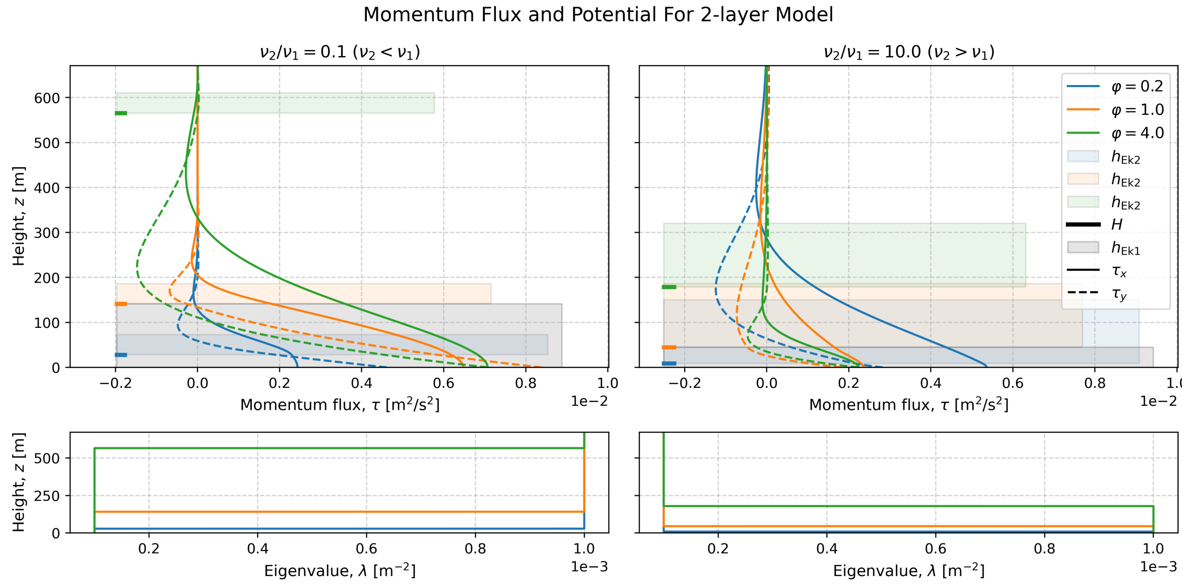

Figure 1: The vertical structure of the momentum flux for the two layer model. The momentum flux is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{Ek1}}$ and the ratio of the viscosities. The corresponding Ekman heights and eigenvalue is shown with the vertical momentum flux. The solution assumed a purely zonal geostrophic flow and a Coriolis parameter of $f = 1 \times 10^{-4}\,\mathrm{s}^{-1}$.

By defining the constant R

$$R = \frac{1-\gamma}{1+\gamma}. \tag{19}$$

and using the derived solution for the vertical momentum flux, the surface wind-turning angle becomes

$$\theta = \arg\left[(A_1 + B_1)\right] \tag{20}$$

$$= 45^\circ + \arg\left[\frac{e^{(1+i)\varphi} - Re^{-(1+i)\varphi}}{e^{(1+i)\varphi} + Re^{-(1+i)\varphi}}\right], \tag{21}$$

which can be observed in Figure 2. There solutions for different viscosity ratios can be observed, half of which have an higher upper-level viscosity and vice versa. One can observe a convergence towards the classical Ekman value of 45° when $\varphi \to \infty$ for all the viscosity ratios. This can be attributed to the high altitude interface H not acting in the domain of the spiral, which reduces the problem to a one layer model.

The zero limit is, however, dependent on the viscosity ratio, where finite ratios approach 45°. This occurs when the interface H is close to the surface, again reducing the problem to a one layer model if $H = 0$. However, when $\frac{\nu_2}{\nu_1} \to 0$ and $\varphi \to 0$ the angle becomes $\theta \to 90^\circ$. Respectively, when $\frac{\nu_2}{\nu_1} \to \infty$ and $\varphi \to 0$, $\theta \to 0^\circ$. This indicates that the interface approaches the ground while the viscosity contrast becomes extreme, leading to a surface-dominated response where the surface stress aligns either perpendicular or parallel to the upper geostrophic flow.

In the case of a very much weaker upper viscosity the perpendicular angle can be interpreted as the entire turning of the Coriolis force has to occur at the surface, since the viscosity becomes irrelevant aloft. In the opposite case, a weaker viscosity at the ground makes the Ekman spiral less prominent. Since the Ekman spiral occurs due to the surface stress, thus, if no major viscosity is present the stress becomes negligible and the wind-turning does not occur. For $\nu_1 > \nu_2$ (and $\nu_2 > \nu_1$ respectively) the solution will always provide a result greater than 45°, except between the $\varphi \in \left(\frac{\pi}{2}, \pi\right)$ were a angle below 45° can be observed.

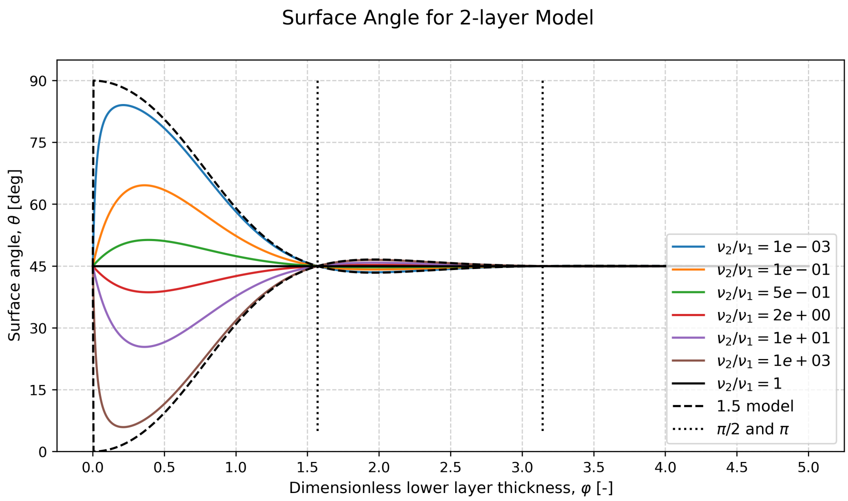

Figure 2: The surface angle as a function of the dimensionless layer thickness $\varphi = H/h_{\text{Ek1}}$ for the two-layer viscosity model. Different coloured curves correspond to different layer viscosity ratios. The classical Ekman angle of 45° is shown as a reference.

2.5 The exponential decaying viscosity

Another viscosity profile that can provide a fully analytical solution is an exponentially decaying viscosity with altitude. Here the viscosity is set to decay from an initial viscosity of ν_0, with a decay rate of k as $\nu(z) = \nu_0 \exp(-kz)$. Solving Equation 9 with this viscosity profile can be done by introducing the new variable

$$t = 4\,(1+i)\;\varphi_{\text{exp}}\, e^{\frac{kz}{4}}. \tag{22}$$

The non-dimensional parameter φ_{exp} resembles the non-dimensional layer thickness from the two-layer model, but defined as

$$\varphi_{\text{exp}} = \frac{1}{k\,h_{\text{Ek0}}} = \frac{1}{k}\sqrt{\frac{f}{2\nu_0}}. \tag{23}$$

Using these newly defined variables Equation 9 transforms into

$$t^2 \frac{\mathrm{d}^2\tau}{\mathrm{d}t^2} + t\frac{\mathrm{d}\tau}{\mathrm{d}t} - t^2\tau = 0, \tag{24}$$

which is a differential equation with a standard solution. The standard solution is a modified Bessel equation as

$$\tau(t) = C_1 I_0(t) + C_2 K_0(t), \tag{25}$$

where C_1 and C_2 are constants, while $I_0(t)$ and $K_0(t)$ are modified Bessel functions of the first and second kind of order zero, respectively. By applying the previously defined boundary conditions from Equation 10 the two constants can be found. Applying the geostrophic boundary condition from Equation 10b shows that $C_1 = 0$ since $I_0(t)$ does not go to zero when $t \to \infty$ (which occurs when $z \to \infty$). Applying the surface boundary condition from Equation 10a an expression for the second constant can be found. By transforming the solution to depend on the variable z gives the solution

$$\tau(z) = \frac{(1+i)\,fU_g}{2k\varphi}\,\frac{K_0\left(4\,[1+i]\,\varphi_{\text{exp}}e^{kz/4}\right)}{K_1\left(4\,[1+i]\,\varphi_{\text{exp}}\right)}. \tag{26}$$

This full solution for the vertical momentum flux in the exponential model can be observed in Figure 1. There, the change of the vertical momentum flux with altitude for different non-dimensional heights, φ, can be observed. For smaller φ, the solution is more localized toward the surface, whereas larger φ produces a more extended spiral. This implies that for larger φ, the viscosity changes very little, causing the spiral to behave as in the one-layer model. In contrast, smaller φ indicates a rapid viscosity decrease from the ground.

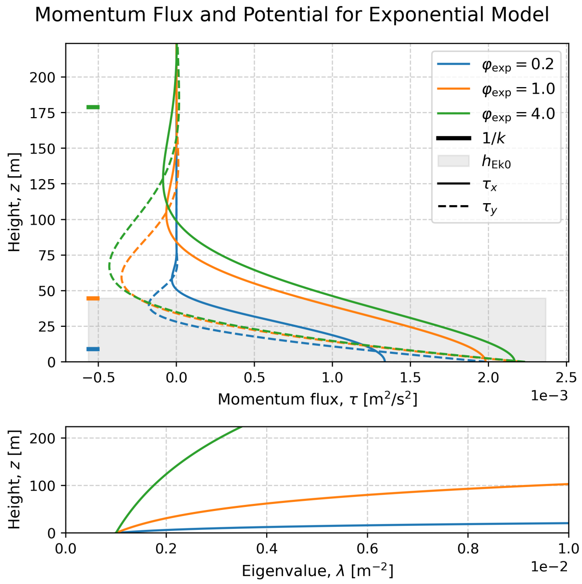

Figure 3: The vertical structure of the momentum flux for exponentially decreasing viscosity. The momentum flux is dependent on the dimensionless parameter $\varphi_{\mathrm{exp}} = 1/k\, h_{\mathrm{Ek0}}$. The corresponding Ekman heights and eigenvalue are shown with the vertical momentum flux. The solution assumed a purely zonal geostrophic flow and a Coriolis parameter of $f = 1 \times 10^{-4}\,\mathrm{s}^{-1}$

Using Equation 26 the surface angle of this model gives

$$\theta = 45^\circ + \arg\left[\frac{K_0\left(4\,[1+i]\;\varphi_{\mathrm{exp}}\right)}{K_1\left(4\,[1+i]\;\varphi_{\mathrm{exp}}\right)}\right], \tag{27}$$

where K_1 is the modified Bessel function of the second kind of order one. Figure 4 showcases the result from the surface wind-turning angle. As this model only provides scenarios when the viscosity increases from the ground, all angels are greater then the reference angle of 45°, as seen by the two layer model. One can observe a similar limit as in Figure 2 when $\varphi \to 0$ for $\nu_2 = 0$, as Equation 27 also approaches 90°. The same limit is also true for $\varphi \to \infty$, which gives $\theta \to 45°$, although this exponential solution converges slightly slower than the two-layer solution. The reasoning behind this convergence is that the lower decay rates makes the solution behave more like the classical Ekman spiral with constant viscosity. The opposite case when the layer is really thin forces a strong turning close to the surface.

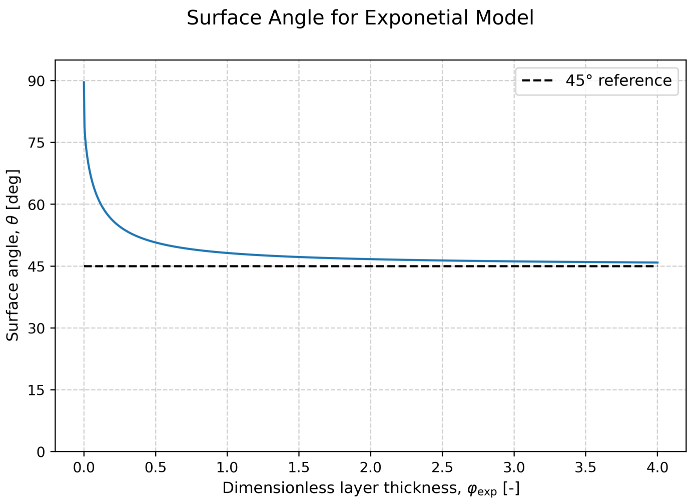

Figure 4: Surface angle for the exponentially decaying viscosity model as a function of the dimensionless parameter $\varphi = H/h_{\mathrm{Ek}}$. The solution assumed a purely zonal geostrophic flow.

2.6 The simplified upper boundary condition model

In the previous sections two completely analytical solutions for two different altitude-varying viscosity profiles were derived. The difficulty in solving Equation 9 comes from the altitude dependence of the viscosity profile $\nu(z)$. This introduces variable coefficients into the complex second-order ordinary differential equation, which in general prevents closed-form analytical solutions. Analytical solutions therefore exist only for specific viscosity profiles that allow the equation to be transformed into a solvable form.

Equation 9 is furthermore also difficult to solve in a numerical setting, as the upper boundary condition is a limit. This will put limits on the viscosity profiles to study, as the viscosity cannot increase with altitude continuously. In the physical world, this makes sense, since we assume that high up in the atmosphere (or deep down in the ocean), the flow becomes laminar and the turbulent viscosity small. This was what was labelled as the geostrophic limit. However, by limiting ourselves to only decreasing viscosity profiles, the real-world differential equation becomes impossible to solve.

A solution to this problem is to introduce the simplified upper boundary condition model. For the two-layer model, this is reduced to a 1.5 layer model, where we assume that the upper-level viscosity is negligible $\nu_2 \approx 0$, which could be observed as the limit in Figure 2. This also makes physical sense, since we know that both the atmosphere and the ocean consist of a boundary layer where turbulent mixing takes place. Allowing the viscosity to vary freely in the lower layer reduces the problem to a simplified upper boundary condition model. This will allow viscosity profiles that do not converge at $z \to \infty$, since the new domain is between $z \in [0, H]$. This modifies the boundary conditions in Equation 10 as

$$\left.\frac{\mathrm{d}\tau}{\mathrm{d}z}\right|_{z=0} = -ifU_g, \tag{28a}$$

$$\tau(z = H) = 0. \tag{28b}$$

Above, H becomes the height of the boundary layer, and the upper boundary condition was argued from the jump condition in the two-layer model, as seen in Equation 16b.

2.6.1 Simple viscosity profiles for the SUBC model

Using the the simplified upper boundary condition, SUBC-, model simple viscosity profiles can be studied and evaluated numerically. The easiest, except for the constant viscosity, is a linearly varying viscosity with height as

$$\nu_\uparrow(z) = \frac{\nu_{\max}}{H} z, \tag{29a}$$

$$\nu_\downarrow(z) = \frac{\nu_{\max}}{H} (H - z), \tag{29b}$$

where $\nu_{\max}$ is the maximal viscosity. For both of these profiles a dimensionless thickness coordinate can be defined as

$$\tilde{z} = \frac{z}{H} \implies \begin{cases} \nu_\uparrow(z) = \nu_{\max}\tilde{z} \\ \nu_\downarrow(z) = \nu_{\max}(1 - \tilde{z}). \end{cases} \tag{30}$$

This new coordinate changes the previous z-derivative, as a factor of $1/H$ appears when taking the derivative with respect to $\tilde{z}$. Using these linear viscosity profiles, together with the new dimensionless thickness coordinate, rewritesEquation 9 for the two profiles as

$$\frac{\mathrm{d}^2\tau}{\mathrm{d}\tilde{z}^2} = i \underbrace{\frac{1}{\tilde{z}}}_{\equiv 1/\tilde{\nu}_\uparrow} \underbrace{\frac{Hf^2}{\nu_{\max}}}_{=2\varphi^2} \tau, \tag{31a}$$

$$\frac{\mathrm{d}^2\tau}{\mathrm{d}\tilde{z}^2} = i \underbrace{\frac{1}{1-\tilde{z}}}_{\equiv 1/\tilde{\nu}_\downarrow} \underbrace{\frac{Hf^2}{\nu_{\max}}}_{=2\varphi^2} \tau. \tag{31b}$$

Above the dimensionless thickness φ was again introduced following the logic from before.

Another simple profile to evaluate is a parabolic increasing viscosity as

$$\nu_p = \frac{4\nu_{\max}}{H^2}\left(H - z\right)z. \tag{32}$$

Again, using the non-dimensional height coordinate $\tilde{z}$ and the dimensionless thickness φ, rewrites Equation 9 as

$$\frac{\mathrm{d}^2\tau}{\mathrm{d}\tilde{z}^2} = i\underbrace{\frac{1}{4\tilde{z}\left[1-\tilde{z}\right]}}_{\equiv 1/\tilde{\nu}_p}\underbrace{\frac{Hf^2}{\nu_{\max}}}_{=2\varphi^2}(U - U_g). \tag{33}$$

Since the height coordinate is normalized for the height H, the dimensionless thickness simply becomes the same as the inverse of the normalized Ekman height as

$$\varphi = \frac{1}{\tilde{h}_{\mathrm{Ek}}}. \tag{34}$$

3 Methodology

3.1 Numerical method

For arbitrary viscosity profiles, Equation 4 generally does not admit a closed-form analytical solution. The equation was therefore solved numerically as a boundary value problem using the `solve_bvp` routine available in `SciPy` [Virtanen et al., 2020].

For the SUBC model, the equation can be rewritten in terms of a dimensionless layer thickness, φ. Using an arbitrary height-normalized eddy viscosity profile, $\tilde{\nu}(z)$, the differential equation simplifies. Separating into zonal and meridional velocity components yields:

$$\frac{\mathrm{d}^2\tau_{\mathrm{x}}}{\mathrm{d}z^2} = -\frac{1}{\tilde{\nu}(z)}2\varphi^2\tau_y, \tag{35a}$$

$$\frac{\mathrm{d}^2\tau_{\mathrm{y}}}{\mathrm{d}z^2} = \frac{1}{\tilde{\nu}(z)}2\varphi^2\tau_x \tag{35b}$$

Here it was also assumed that the wind aloft was set as a purely zonal u_g. To apply the numerical boundary value solver, the equations were rewritten as a first-order system by introducing the new variables

$$\tau_x' = \frac{\mathrm{d}\tau_x}{\mathrm{d}z}, \qquad \tau_y' = \frac{\mathrm{d}\tau_y}{\mathrm{d}z}. \tag{36}$$

Expanding Equation 35b into four coupled first-order equations gives

$$\begin{cases} \frac{\mathrm{d}\tau_x}{\mathrm{d}z} &= \tau_x', \\ \frac{\mathrm{d}\tau_x'}{\mathrm{d}z} &= -\frac{2\varphi^2\tau_y}{\tilde{\nu}(z)}, \\ \frac{\mathrm{d}\tau_y}{\mathrm{d}z} &= \tau_y', \\ \frac{\mathrm{d}\tau_y'}{\mathrm{d}z} &= \frac{2\varphi^2\tau_x}{\tilde{\nu}(z)} \end{cases} \tag{37}$$

with the boundary conditions

$$\tau_x'(\varphi = 0) = 0, \qquad \tau_x(\varphi = 1) = 0, \tag{38a}$$

$$\tau_y'(\varphi = 0) = -fu_g, \qquad \tau_y(\varphi = 1) = 0. \tag{38b}$$

If the viscosity profile, $\tilde{\nu}$, obtains a value of zero in the domain, then the coupled Equation 37 becomes unsolvable since we divide by $\hat{\nu} = 0$. In this scenario one has to introduce a constant background viscosity, ϵ as

$$\hat{\nu} \to \hat{\nu} + \epsilon, \quad \text{where } \epsilon = \frac{\nu_{\mathrm{background}}}{\nu_{\max}} \tag{39}$$

and $\hat{\nu}/\epsilon \gg 1$ was assumed. This background viscosity was thus applied for the linear decreasing, linear increasing, and the parabolic model since they have zero viscosity at the surface or at the top.

The SUBC model equations were solved iteratively on a vertical grid. The solver required an initial guess, which was chosen based on the initial setup of the system. The initial guess was therefore set as the classical solution for a constant viscosity. The final solution was found when the iterative solution reached a steady state for the differential equations with the imposed boundary conditions.

4 Result

4.1 The vertical structure of the momentum flux

The vertical profiles from the numerical SUBC model can be viewed in Figure 5–7. One can view this together with the normalized Ekman height for the different layer. An important note is that when $\varphi < 1$ then the Ekman heights become larger than the limit H. This is the same as stating that the entire spiral occurs really close to the surface.

Since the surface wind-turning angle is calculated by comparing the initial change of the viscosity, both the linear increasing and the parabolic models are dependent on the background viscosity ϵ. The effect of varying ϵ can be observed in Figure 6–7.

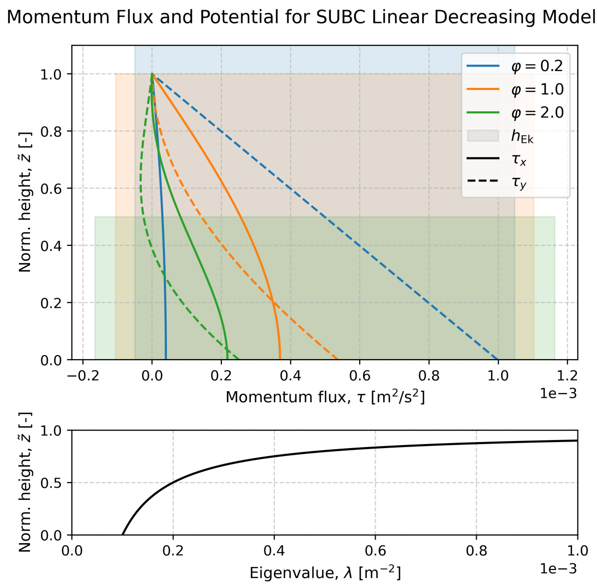

Figure 5: Numerically computed vertical structure for linearly decreasing viscosity. The momentum flux is dependent on the dimensionless parameter $\varphi = \tilde{h}_{\text{Ek}}^{-1}$. The corresponding Ekman heights and eigenvalue is shown with the shown with the vertical momentum flux. The solution assumed a purely zonal geostrophic flow.

4.2 The surface angles

Figure 8–10 show the surface angles from the numerical SUBC models. Similarities between these models, the increasing two-layer model, and the exponentially decaying model can be observed. They all have a lower boundary of 90° and an upper boundary of 45°. The reason for the similarly between the exponential model and the increasing two layer model, is that the SUBC model is based on the increasing two layer model.

Thus, in the limit when $\varphi \to 0$ the Ekman height becomes infinitely larger then the interface H, providing a 1.5 layer like solution which has the the limit of 90°. When $\varphi \to \infty$ the maximum value of the profile decreases, providing a more constant like solution, with a smaller viscosity change with altitude. Thus the solution approaches the classical Ekman solution, with an angle of 45°.

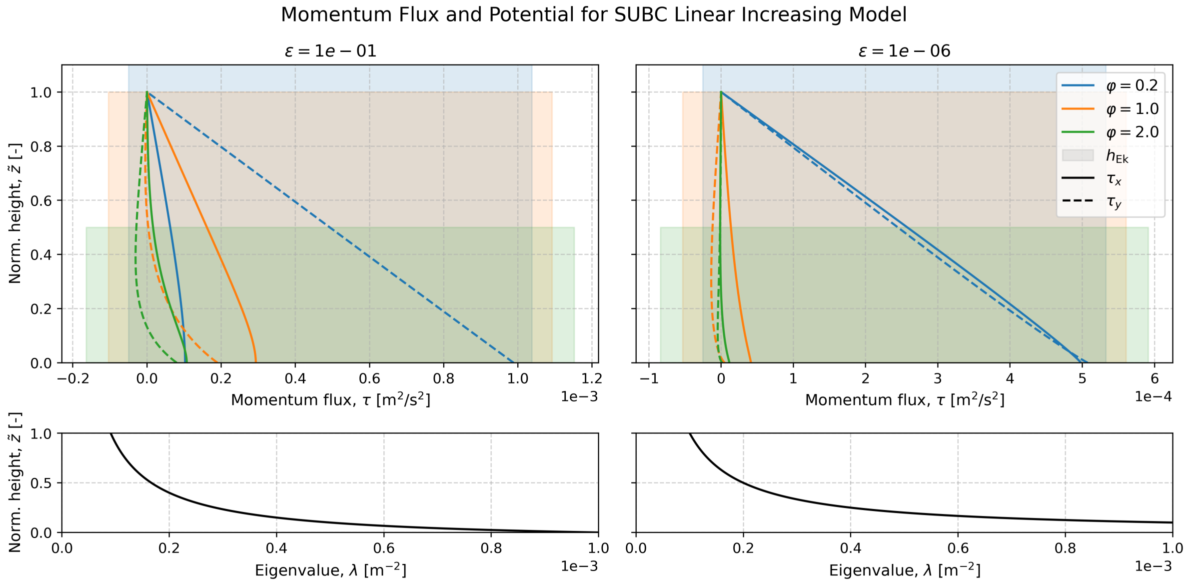

Figure 6: Numerically computed vertical structure for linearly increasing viscosity. The momentum flux is dependent on the dimensionless parameter $\varphi = \tilde{h}_{\mathrm{Ek}}^{-1}$ and the background viscosity ϵ. The corresponding Ekman heights and eigenvalue is shown with the shown with the vertical momentum flux. The solution assumed a purely zonal geostrophic flow.

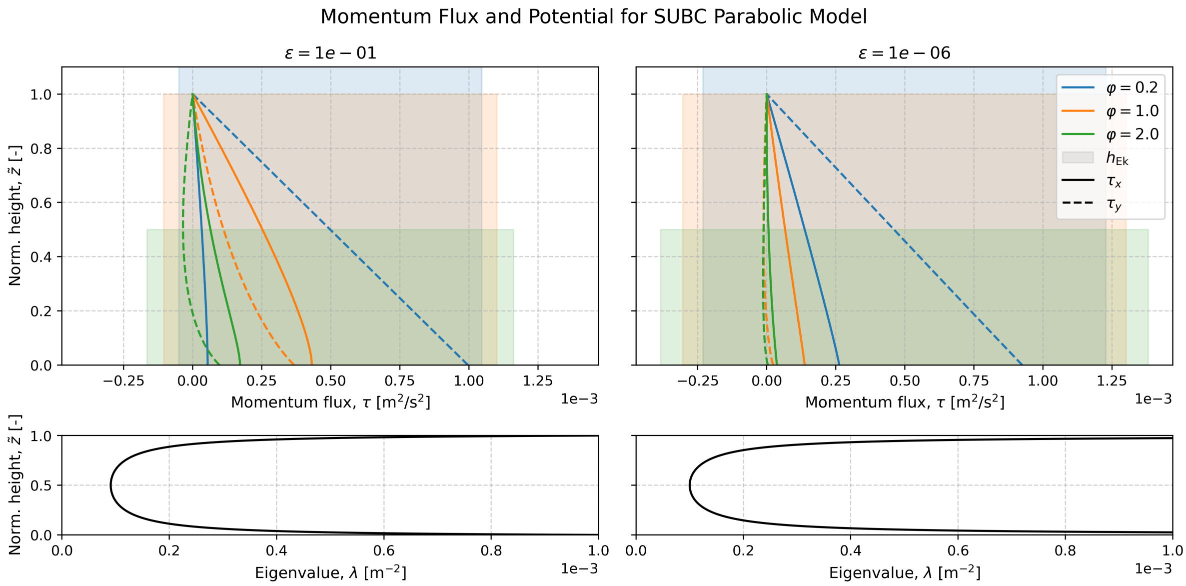

Figure 7: Numerically computed vertical structure for parabolic viscosity. The momentum flux is dependent on the dimensionless parameter $\varphi = \tilde{h}_{\mathrm{Ek}}^{-1}$ and the background viscosity ϵ. The corresponding Ekman heights and eigenvalue is shown with the shown with the vertical momentum flux. The solution assumed a purely zonal geostrophic flow.

The result from the linearly decreasing SUBC model is shown in Figure 8, where the values smoothly transition from $90° \to 45°$ as φ increases. This plot resembles the exponential decaying model seen in Figure 4 and the 1.5 model as seen in Figure 2. Again, all angles are larger then the reference angle of 45°. The close resemblance to the exponential and 1.5 layer solutions makes sense since this solution is also a strictly decreasing solution.

For the linearly increasing viscosity a different result can be seen in Figure 9. The upper boundary is again located at 45°, but at a very large non-dimensional height ratio. For smaller background viscosities, ϵ, this limit occurs at many magnitudes higher then in the previous models. Since the eigenvalue is proportional to the inverse o the viscosity, smaller background viscosities yields way larger eigenvalues as seen in

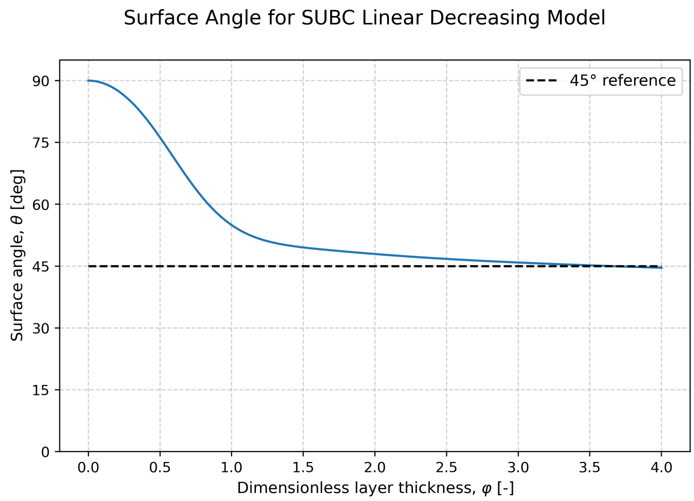

Figure 8: Numerically computed surface angles for decreasing viscosity profiles in the SUBC model. The angle is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{Ek}}$. The solution assumed a purely zonal geostrophic flow.

Figure 6. Thus, this limit of 45° occurs at way higher non-dimensional heights.

The curve also the obtains values smaller then 45° outside of the limits discussed above. Since the limits are the extreme cases, this interior is the more interesting part of the result. The surface wind-turning angle is smaller then 45° since a larger viscosity ata a higher altitude means that the turning of the wind is smaller near the surface and greater aloft. This forces the surface angles to become less then 45° near the ground.

Another interesting observation is that, away from the limits, she surface wind-turning angle is more dependent on the background viscosity, ϵ, rather then the non-dimensional height, especially for the lower ϵ. This occurs since the eigenvalue is the inverse of the viscosity and that the initial increase from the ground to the domain is the most crucial one for the surface angle. Thus, if the background viscosity is much smaller then the maximal viscosity then smaller ϵ is obtained and the initial eigenvalue jump from the surface becomes greater. In the theoretical limit when $\epsilon \to 0$, the initial eigenvalue jump would be infinite, and the surface angle would be zero.

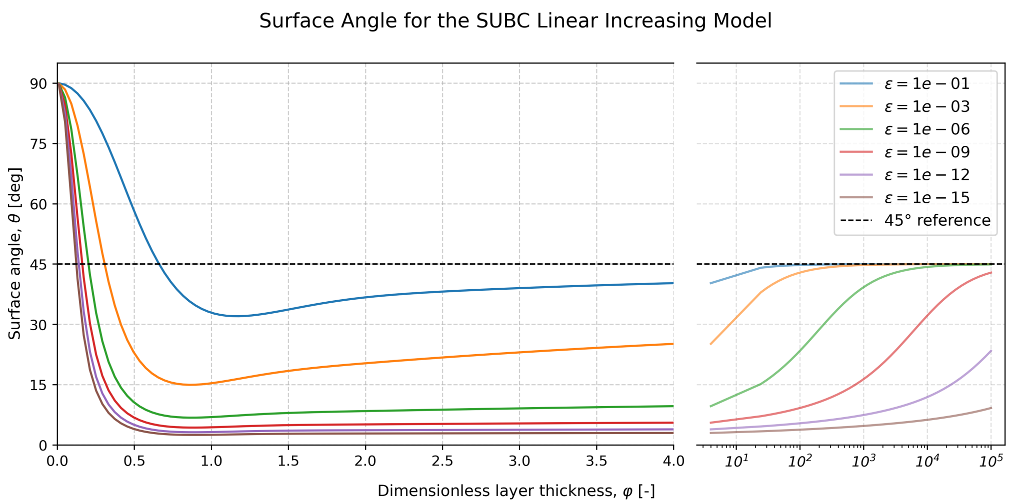

Figure 9: Numerically computed surface angles for linearly increasing. The angle is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{Ek}}$ and the background viscosity ϵ. The solution assumed a purely zonal geostrophic flow.

The SUBC model with a parabolic viscosity profile gave Figure 10. Very similar behaviour to the increasing viscosity profile in Figure 9 can be observed. Since the viscosity jump from the surface is what determines the surface angle, the parabolic decrees aloft has no dramatic effect on the surface angle. The two models differ slightly in that the parabolic curves cross the 45° later than the linearly increasing curves.

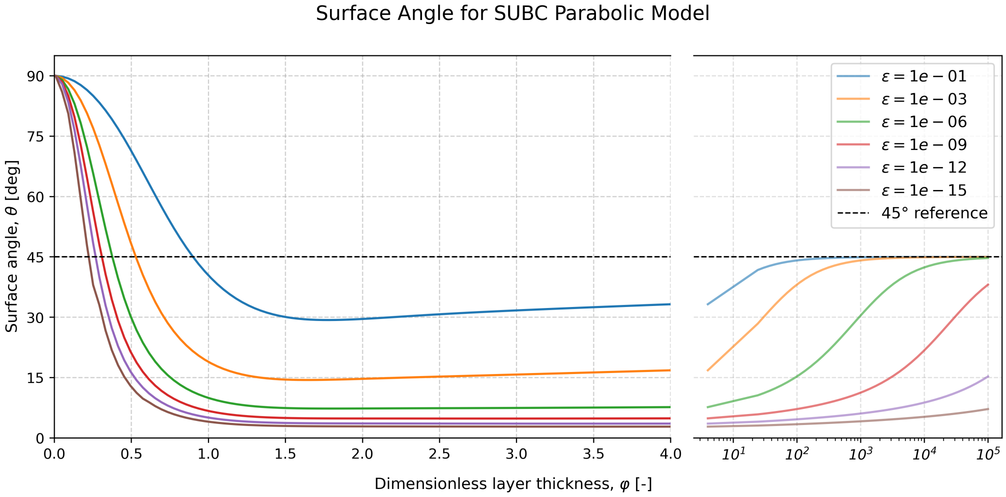

Figure 10: Numerically computed surface angles for a parabolic viscosity profile in the SUBC model. The angle is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{Ek}}$ and the background viscosity ϵ. The solution assumed a purely zonal geostrophic flow.

4.2.1 The Ekman transport

Computing the transport as by the surface vertical momentum flux as Equation 12 yielded exactly the same shapes as seen in Figure 2–10, but centerd around 135° rather then 45°. The angle differences between the surface angles and the transport angle was thus always 90°, just like in the classical Ekman theory.

5 Discussion

6 Conclusion

A The Source Code

The source code for this article can be found on GitHub under:

`https://github.com/FredrikBergelv/Ekman-Spiral-Project`

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